GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

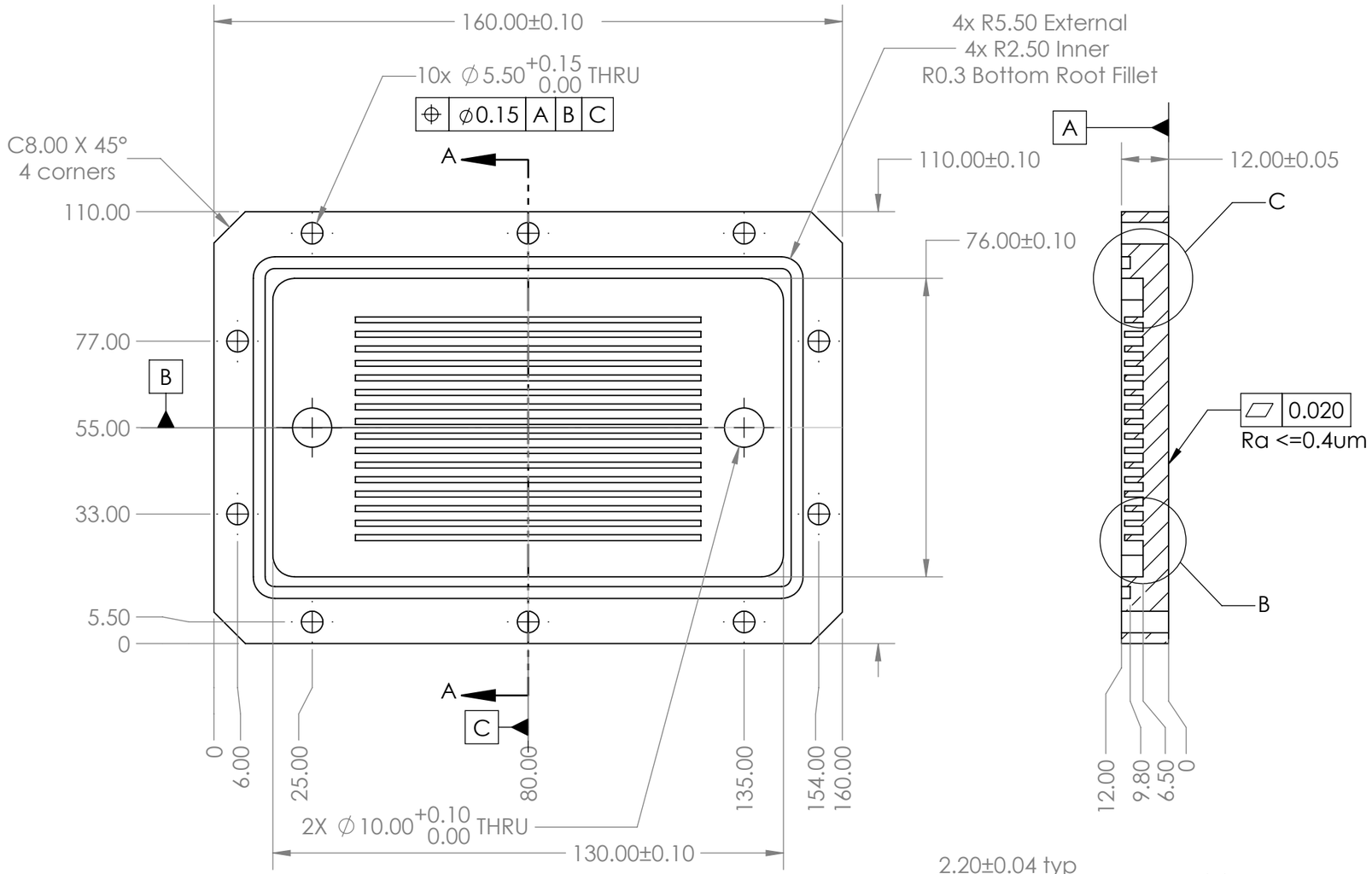
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-AUTO-COOLING-BASE-4.1.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/- 0.10 mm, ANGULAR +/- 0.2 DEG (CNC PRECISION MILLING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.

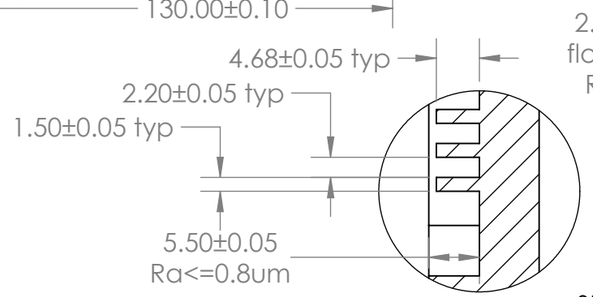
2. MATERIAL & THERMAL PERFORMANCE SPECIFICATIONS:
2.1 MATERIAL: AUTOMOTIVE GRADE WROUGHT ALUMINUM ALLOY 6061-T6 / AlSi10Mg.
2.2 THERMAL CONDUCTIVITY: k >= 170 W/(m·K) AT 25 DEG C.
2.3 100% SUBTRACTIVE 3-AXIS / 5-AXIS CNC HIGH-SPEED MACHINING ONLY. NO ADDITIVE METAL 3D PRINTING PERMITTED TO PREVENT COOLANT RESIDUAL PARTICULATE SHEDDING AND VALVE CLOGGING.

3. GEOMETRY & GD&T CONTROLS:
3.1 PRIMARY DATUM [A] (SiC MODULE THERMAL MOUNTING FACE): FLATNESS <= 0.020 mm, SURFACE FINISH Ra <= 0.4 MICRONS.
3.2 MOUNTING BORES (10X M5): TRUE POSITION WITHIN 0.15 mm RELATIVE TO [A][B][C]. ZERO INTERFERENCE WITH O-RING SEAL ISOLATION WALL (MINIMUM WALL THICKNESS >= 1.50 mm MANDATORY).
3.3 O-RING SEAL GROOVE: WIDTH 3.00 +0.08/-0.00 mm, DEPTH 2.20 +/- 0.04 mm, CONTINUOUS AND CHATTER-FREE (Ra <= 0.8 MICRONS).
3.4 INTERNAL COOLING FINS: 100% 2.5D VERTICAL MACHINABILITY (UNDERCUT-FREE) WITH UNIFORM FIN THICKNESS 1.50 mm AND INTER-RIB GAP 2.20 mm.

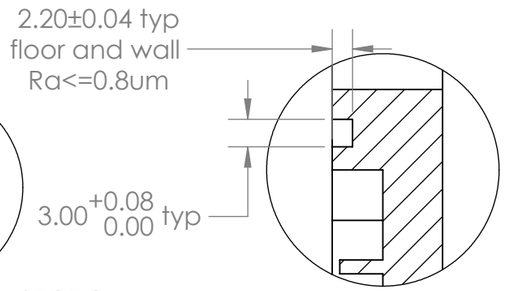
4. PRESSURE TESTING & LEAKAGE INTEGRITY:
4.1 HYDROSTATIC PRESSURE TEST: 100% AIR-UNDER-WATER / HELIUM PROOF TEST AT 6.0 BAR (0.6 MPa) FOR 60 SECONDS. ZERO LEAKAGE PERMITTED.
4.2 PRESSURE DROP REQUIREMENT: PRESSURE DROP (DELTA P) ACROSS INLET/OUTLET <= 25 kPa AT RATED FLOW RATE OF 10 L/min (50/50 WATER-GLYCOL MIXTURE).

5. SURFACE TREATMENT & CLEANLINESS:
5.1 SURFACE FINISH: MIL-A-8625 TYPE III HARD ANODIZED (25-35 MICRONS COATING) FOR CORROSION INHIBITION, WITH BOTTOM DATUM [A] THERMAL ISLAND MASKED (BARE CONDUCTIVE ALUMINUM FOR THERMAL GREASE COUPLING).
5.2 CLEANLINESS PROTOCOL: MULTI-STAGE ULTRASONIC CAVITATION CLEANING TO REMOVE RESIDUAL CUTTING FLUIDS AND SWARF. COMPLIANT WITH AUTOMOTIVE ISO 16232 PARTICULATE EXTRACTION STANDARDS.





SECTION A-A



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH: Hard Anodized Mil-A-8625 Type III (Non-Sealed on Thermal Island), Electroless Nickel Plating Optional		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE			TITLE: EV INVERTER SiC POWER MODULE LIQUID COOLING BASE PLATE	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL: Automotive / Aerospace Grade Aluminum Alloy 6061-T6 / AlSi10Mg		DWG NO. W3-AUTO-COOLING-BASE-4.1	A3
				WEIGHT: ~348.5 g		SCALE: NTS	SHEET 1 OF 1